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Erkan Tairi

Education

2019–2024 **PhD in Computer Science**, *TU Wien*, Vienna, Austria
Supervisors: Matteo Maffei (TU Wien) and Daniel Slamanig (AIT Austrian Institute of Technology)
Thesis: Foundations of Adaptor Signatures for Distributed Ledger Protocols
`https://doi.org/10.34726/hss.2024.123264`

2016–2018 **MSc in Computer Science**, *Johannes Kepler University*, Linz, Austria

2011–2015 **BSc in Computer Science**, *University St. Paul the Apostle*, Ohrid, Macedonia

Professional Experience

Oct. 2025 - **Postdoctoral Researcher**, *UC Berkeley*, Berkeley, CA, USA
Host: Sanjam Garg

Sep. 2025 - **Research Scientist**, *Alloc Init*, Remote
Cryptographic protocol design and documentation

May 2024 - Sep. 2025 **Postdoctoral Researcher**, *ENS Paris*, Paris, France
Host: David Pointcheval

Feb. 2022 - Jan. 2024 **Research Engineer**, *MyPrivacy*, Vienna, Austria
Cryptographic protocol design and development

Internships and Visits

Oct. 2026 **Research Visit**, *Tsinghua University*, Beijing, China
Host: Yang Yu

May 2026 **Research Visit**, *ENS Paris*, Paris, France
Host: Brice Minaud

Feb. 2025 **Research Visit**, *King's College London*, London, England
Host: Martin Albrecht

Dec. 2022 **Research Visit**, *IMDEA Software Institute*, Madrid, Spain
Host: Dario Fiore

Nov. 2022 **Research Visit**, *ETH Zürich*, Zürich, Switzerland
Host: Dennis Hofheinz

Jun.-Sep. 2018 **Internship**, *AIT Austrian Institute of Technology*, Vienna, Austria
Supervisor: Daniel Slamanig

Oct. 2018 - Mar. 2019 **Internship**, *TU Wien*, Vienna, Austria
Supervisor: Matteo Maffei

Teaching and Supervision

2019–2023 **Teaching Assistant**, *TU Wien*, Vienna, Austria
Introduction to Cryptography, Privacy-Enhancing Cryptography, Cryptocurrencies, (graduate level)

Master Thesis Supervision
Aron Wussler (TU Wien) – Post-quantum Cryptography in OpenPGP
`https://doi.org/10.34726/hss.2023.106226`
Ngaffo Raissa Flora (University of Bamenda) – Simulation-extractable NIZK Proof Systems

Publications

Published Papers

Registered Functional Encryption for Pseudorandom Functionalities from Lattices: Registered ABE for Unbounded Depth Circuits and Turing Machines, and More. In IACR Communications in Cryptology (CiC).
Tapas Pal, Robert Schädlich and Erkan Tairi
https://eprint.iacr.org/2025/967

Lighthouse: Single-Server Secure Aggregation with O(1) Server-Committee Communication. In USENIX Security 2026.
Sanjam Garg, Alireza Kavousi, Dimitris Kolonelos, Erkan Tairi, and Zhipeng Wang
https://eprint.iacr.org/2026/346

LeOPaRd: Towards Practical Post-Quantum Oblivious PRFs via 2HashDH Paradigm. In CRYPTO 2026.
Muhammed F. Esgin, Ron Steinfeld, Erkan Tairi and Jie Xu
https://eprint.iacr.org/2024/1615

(Fine-Grained) Unbounded Inner-Product Functional Encryption from LWE. In SCN 2026.
Valerio Cini and Erkan Tairi
https://eprint.iacr.org/2026/062

Ciphertext-Updatable Attribute-based and Predicate Encryption from Lattices. In SCN 2026.
Robert Schädlich, Linda Scheu-Hachtel, Erkan Tairi and Yuejun Wang

Hardness of M-LWE with General Distributions and Applications to Leaky Variants. In PKC 2026.
Katharina Boudgoust, Corentin Jeudy, Erkan Tairi and Weiqiang Wen
https://eprint.iacr.org/2025/1472

On Verifiable Delay Functions from Time-lock Puzzles. In PKC 2026.
Hamza Abusalah, Karen Azari, Dario Fiore, Chethan Kamath and Erkan Tairi
https://eprint.iacr.org/2025/1782

Impossibility of VDFs in the ROM: The Complete Picture. In EUROCRYPT 2026.
Hamza Abusalah, Karen Azari, Chethan Kamath, Erkan Tairi and Maximilian von Consbruch
https://eprint.iacr.org/2025/1773

Lower Bounds for Lattice-based Compact Functional Encryption. In EUROCRYPT 2024.
Erkan Tairi and Akin Ünal
https://doi.org/10.1007/978-3-031-58723-8_9

(Inner-Product) Functional Encryption with Updatable Ciphertexts. In Journal of Cryptology.
Valerio Cini, Sebastian Ramacher, Daniel Slamanig, Christoph Striecks and Erkan Tairi
https://doi.org/10.1007/s00145-023-09486-y

Optimizing 0-RTT Key Exchange with Full Forward Security. In ACM CCSW 2023.
Christian Göth, Sebastian Ramacher, Daniel Slamanig, Christoph Striecks, Erkan Tairi and Alexander Zikulnig
https://doi.org/10.1145/3605763.3625246

LedgerLocks: A Security Framework for Blockchain Protocols Based on Adaptor Signatures. In ACM CCS 2023.
Erkan Tairi, Pedro Moreno-Sanchez and Clara Schneidewind
https://doi.org/10.1145/3576915.3623149

Foundations of Coin Mixing Services. In ACM CCS 2022.
Noemi Glaeser, Matteo Maffei, Giulio Malavolta, Pedro Moreno-Sanchez, Erkan Tairi and Sri AravindaKrishnan Thyagarajan
https://doi.org/10.1145/3548606.3560637

A2L: Anonymous Atomic Locks for Scalability in Payment Channel Hubs. In IEEE S&P 2021.
Erkan Tairi, Pedro Moreno-Sanchez and Matteo Maffei
https://doi.org/10.1109/sp40001.2021.00111

Post-Quantum Adaptor Signature for Privacy-Preserving Off-Chain Payments. In FC 2021.
Erkan Tairi, Pedro Moreno-Sanchez and Matteo Maffei
https://doi.org/10.1007/978-3-662-64331-0_7

Updatable Signatures and Message Authentication Codes. In PKC 2021.
Valerio Cini, Sebastian Ramacher, Daniel Slamanig, Christoph Striecks and Erkan Tairi
https://doi.org/10.1007/978-3-030-75245-3_25

Preprints (Under Submission)

Gaussian Kernel Lattices and Smoothing Bounds from Theta Integrals.
Samed Düzlü, Nihar Gargava and Erkan Tairi
https://eprint.iacr.org/2026/2180

Algebraic Cryptanalysis of Hinted vSIS and Decomposed LWE.
Valerio Cini, Robert Schädlich and Erkan Tairi

Lower Bounds for Linear Secret Sharing with Bounded Reconstruction Coefficients.
Yiping Ma and Erkan Tairi

Decentralized Multi-Authority (Attribute-Based) Traitor Tracing.
Pratish Datta, Robert Schädlich and Erkan Tairi
https://eprint.iacr.org/2026/1268

Adaptor Signatures Meet BLS: Efficient Blockchain Applications with Unique Adaptor Signatures.
Javier Gomez-Martinez, Erkan Tairi, Pedro Moreno-Sanchez and Clara Schneidewind
https://eprint.iacr.org/2026/1394

Foundations of Verifiably Encrypted (Blind) Signatures.
Diego Castejon-Molina, Erkan Tairi, Dimitrios Vasilopoulos and Pedro Moreno-Sanchez
https://eprint.iacr.org/2026/789

Professional Activities

Program Committee

USENIX Security 2027, ACM CCS 2025-2026, ACM AsiaCCS 2026, FC 2025-2026, IACR Communications in Cryptology 2025-2026, CVC 2025

External Reviewer

ACISP 2024; ACM CCS 2021-2023; ACM AFT 2022; ACNS 2024-2025; APKC 2021-2022; Asiacrypt 2021, 2024-2025; CANS 2022; Crypto 2023-2026; Eurocrypt 2023, 2025-2026; FC 2021-2024; IEEE S&P 2024; IWSEC 2021-2023; PKC 2026-2027; ProvSec 2020-2023

Administration and Organization

- Co-organizer of ViSP Cryptography Research Meetup
- Member of NDSS 2023 Student Support Committee

Awards and Grants

Erwin Schrödinger Fellowship (by Austrian Science Fund)
https://doi.org/10.55776/J4879

France-Berkeley Fund

Input | Output Research Grant

Internet Freedom Fund (by Open Technology Fund)

(Selected) Presentations and Invited Talks

Lighthouse: Single-Server Secure Aggregation with O(1) Server-Committee Communication
Privacy-Preserving Machine Learning Workshop. Aug. 2026
Bay Area Crypto Day. May 2026

(Fine-Grained) Unbounded Inner-Product Functional Encryption from LWE
Crypto Day at Télécom Paris. Jun. 2025
`https://shorturl.at/DrJEr`
Guest talk at King's College London Cybersecurity Group. Feb. 2025

LedgerLocks: A Security Framework for Blockchain Protocols Based on Adaptor Signatures
IOG Seminar. Nov. 2023

(Inner-Product) Functional Encryption with Updatable Ciphertexts
PICOCRYPT Seminar at IMDEA Software Institute. Dec. 2022
`https://shorturl.at/5RP3o`

A2L: Anonymous Atomic Locks for Scalability in Payment Channel Hubs
IEEE Symposium on Security and Privacy. May 2021
`https://www.youtube.com/watch?v=RPj8efQteyg`

Post-Quantum Adaptor Signature for Privacy-Preserving Off-Chain Payments
Financial Cryptography and Data Security 2021. Mar. 2021
`https://www.youtube.com/watch?v=BqMDnWFoGL0`
Decrypto Seminar. Dec. 2020

Updatable Signatures and Message Authentication Codes
Conference PKC 2021. May 2021
`https://www.youtube.com/watch?v=RqAlXixDZ1M`
Young Researcher Crypto Seminar. May 2021

Computer Skills

Programming	C, C++, Rust *(advanced)*, C#, Python, Shell *(average)*, Java, Go *(basics)*
Web	HTML, CSS, JavaScript (React, Node.js), ASP.NET Web *(advanced)*
Software	Visual Studio, SageMath, Magma, Git, LaTeX
Other	Microsoft Azure, Amazon AWS (cloud computing)

Languages

Macedonian	Native	
English	Fluent	*professional working proficiency*
Turkish	Fluent	*professional working proficiency*
French	Intermediate	*elementary proficiency*
German	Intermediate	*elementary proficiency*
Albanian	Intermediate	*elementary proficiency*

Referees

Sanjam Garg	*sanjamg@berkeley.edu*
Daniel Slamanig	*daniel.slamanig@unibw.de*
David Pointcheval	*david.pointcheval@ens.fr*
Matteo Maffei	*matteo.maffei@tuwien.ac.at*